AI-native industrial production: VC primer

Version 2, dated 2026-10-26 (America/Chicago). Overwritten on each rerun. Built from the anip repo: sources/ (1,437 records across all types), companies/, insights/ (signals, questions, thesis, weekly/2026-W43, producers-comparison, software-first-platforms, cross-sector-map, raising-now, repeat-orders-evidence, unit-economics, failures-and-pivots, incumbents-vs-ai-native, materials-science-semis, must-read-sources, huddle-prep). Free sources only; no AlphaSense searches run by the agent (the AlphaSense-arranged expert calls Lilly filed are used).

New to this topic? Start with section 1 and section 1A (the macro primer), then come back to section 0. Acronyms are spelled out on first use and collected in Appendix A.

How to read this doc. Three kinds of statement are kept apart: - Plain sentence ending in `[src-…]` = sourced fact (a filing, release, dataset or reported figure). - **Claim (who):** = something a company, founder or expert asserted. Sourced, but not verified. - `[inference]` = my read. Not sourced; argue with it.

Confidence (`high/med/low`, per taxonomy) is marked on numbers from a deleted post, Wikipedia, Harmonic estimate or a single press article. Contradictions are shown side by side. Excluded from the category by Lilly and not covered except as failure examples: microagi, Pelico, C-Infinity, Dirac, Synera, Bright Machines (failure example only), Retrocausal, DEEP.FINE.

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0. Decision summary

In one line: the category is real, buyers pay for speed rather than lower prices, and nobody has yet shown that its software makes each part cheaper or better over time.

Verdict [inference]. Back capital-light owner-operators at Series B only where speed and qualification are already paid for (Senra, Freeform, Isembard, as diligence hypotheses). Use Hadrian, Divergent and SendCutSend as benchmarks. Do not pay software multiples for job-shop economics until a company shows unit cost or quality improving with volume. Nobody in the repo shows that yet.

What the full read (28 expert calls, 40 podcasts, cited and uncited sources) changed.

1. **Price is not the wedge.** The only “cheaper” report (Raytheon, about 50% lower) came from RFQ (request for quote) prices before any part was delivered, on a relationship brokered by RTX Ventures, a Hadrian investor [src-20250708-2aa3ad]. A Boeing customer says Xometry is cheapest and a 20-25% price gap means re-sourcing [src-20260802-ef6506]. Three Boeing voices say parity or more: “we have actually paid probably more” (Boeing Defense head of product) [src-20260818-9b2d5c]; Hadrian priced at “PCC or slightly more” (PCC = Precision Castparts, a large incumbent supplier owned by Berkshire Hathaway) (Boeing strategy/M&A lead) [src-20250712-e5e110]. A Northrop GM: “Cost is secondary. Cost is driven usually by schedule” [src-20250707-3b7a46]. The premium buyers pay is for speed.

2. **Quality at complexity is the open risk.** “None of their parts right now are flying” (Boeing Defense) [src-20260818-9b2d5c]. “We haven’t received parts back that matched the specification” (PI, ex-Anduril hardware) [src-20250709-43486e].

3. **Share per customer is capped by design.** SpaceX caps a shop at about 1/3 of its revenue (ex-Hadrian head of engineering) [src-20250708-342eaf]. After a PCC factory fire, Boeing “will

never” concentrate one part family on one supplier [src-20260818-9b2d5c]. Revenue per customer has a ceiling, so growth needs many primes [inference].

4. **The moat is workflow plus qualification, and AI CAM is coming from below.** Insiders say the edge is Opus, Hadrian’s MES (manufacturing execution system: the software that directs work on the shop floor), plus factory design, not AI that writes toolpaths [src-20260817-8867f2] [src-20261004-9b47f5]. CloudNC sells CAM Assist (AI for CAM, the software that turns a 3D model into machine instructions) to every shop and tests it in its own UK factory [src-20261009-7cafd0]. Programs last about 50 years, and suppliers for a 2035 aircraft are being chosen now [src-20260818-9b2d5c].

5. **Materials-rd budgets are not growing fast.** A former Moderna director says buyer budgets have not grown “crazily” in 5-6 years and that the gain is a “step one productivity boost” [src-20260909-af3e60] (low). An Albert Invent director says chemical majors reject royalty models (“The short answer is no”) [src-20260913-3ad55e].

6. **Fast prototype contracts hide the quality gap.** OTA contracts (Other Transaction Agreements, fast prototype contracts outside normal federal rules) waive many mil-spec (military specification) quality requirements; when programs move to standard production contracts, “there’s a gap” in statistical lot inspection (sampling batches of parts to prove they meet spec) that Hadrian “is still maturing” on. At thousands of parts its cost curve “asymptotes at a higher level” than traditional shops, and schedule is “near equal” once full inspection is counted (Boeing executive director, customer spending “low single millions” a year, “a fraction of a percent” of outside spend) [src-20260908-ed94f0]. He thinks “low tens of millions” could replicate Hadrian; a Northrop customer engineer thinks it is hard to copy [src-20260802-36d46c].

7. **Distributed networks win on urgency, not cost.** An Isembard customer paid $2,300-3,000 for 4-5 simple parts in 2-3 days that would cost about GBP150-200 in China, and sends larger batches to Chinese shops [src-20260621-f96bbd]. An Airbus Defence supply-chain head sees such networks limited to prototypes and non-critical parts while primes consolidate to fewer tier-1s [src-20251106-dfddcb].

8. **The only learning-over-time number.** Hadrian’s first programming head: “within a couple of years, we saw a 50% cut” in order-to-first-article time; 70% 24-hour utilization is “very, very realistic”; pricing targets about 30% contribution margin; Opus “is replicable” and the moat is “financial power” plus a five-year head start [src-20261004-9b47f5].

9. **Hadrian’s own numbers are claims, and they cut both ways. Claim (Chris Power, Hadrian CEO, on an investor-hosted panel):** “close to a billion in revenue next year”, about 30% “pure software”, utilization 65-80% vs about 10% at primes, “$4 of revenue on $1 of capex”, labor “five to eight X more efficient … it’s not about being cheaper” [src-20260610-779f8b]. **Claim (Power):** “0% chance of growing revenue unless you spend a billion in CapEx” [src-20260921-dad375]. Read together: capex comes before contracts, and the software share is a stated plan, not a reported number [inference].

10. **High utilization does not need robots. Claim (Jim Belosic, SendCutSend CEO):** lasers run at 86% versus the 77% Amada promised with sheet-loader automation, and top Amada shops are “really happy if they get 50 or 60”; he rejected $4M of conveyance because high-mix, low-volume work defeats rigid automation [src-20260626-541ac8]. The edge is software-driven scheduling and quoting, not robotics [inference].

11. **Roll-up economics set the floor. Claim (Re:Build co-founder):** “effectively it’s a 30% gross margin”, acquisitions at “seven times EBITDA” [src-20241001-28c2aa]. That is the incumbent benchmark an AI-native plant must beat to justify venture capital [inference].

12. **Policy now buys demand, not just capacity.** The FY26 NDAA authorizes multiyear munitions procurement (Sec. 804(b)), a purchase-commitment program with advance payments (Sec. 867) and a spares review aimed at advanced manufacturing (Sec. 1842) [src-20251218-f0d6ef]. These are the first channels that could turn prototype wins into multi-year orders [inference].

Evidence on the four thesis claims (claims as numbered in insights/thesis.md)

#	Thesis claim	For	Against	Read [inference]
1	Demand for domestic production is structural	Lockheed 7-year cell deal [src-20261014-015327]; Navy $900M [src-20260320-67ba4c]; Thales CPO: capacity is the pain after COVID-era supplier bankruptcies [src-20251110-84b67e]; Boeing has used parts taken from non-flying aircraft [src-20260818-9b2d5c]	34% spindle utilization at existing shops [src-20250120-419768]; Taiwan 20-30% cheaper and Vietnam/Malaysia cheaper still [src-20250709-43486e]; Boeing makes about 40% of parts in-house and holds 6 months of inventory [src-20250712-e5e110]	True for defense capacity and speed. Commercial rate demand not shown
2	Owning the plant creates a data loop	Freeform and Mariana score 2 (section 6). Ex-Hadrian: about 80% of non-CAM labor automated, lead times cut about 50% [src-20250708-342eaf]; reprogramming saved 50-80% of CAM time [src-20260611-7b80f0]	Toolpaths come from third-party CAM, with Vericut (simulation software) as the check [src-20260611-7b80f0] [src-20261004-9b47f5]; CMM (coordinate measuring machine) inspection is manual [src-20260817-8867f2]; a buyer asks “is it actually” software-enabled [src-20250729-a0a78c]	Workflow automation is real. Learning with volume is not evidenced
3	Moats: process data, secured capacity, qualification	Trusting a new supplier takes 5-10 years [src-20250712-e5e110]; change-of-control clauses in defense contracts (contracts that can be voided if the supplier is sold) [src-20260817-8867f2]; lock-in from 50-year programs [src-20260818-9b2d5c]	Share caps (finding 3); 90% of the first 10 Hadrian employees gone by year 5 [src-20260817-8867f2]; incumbent PCC is also automating [src-20250712-e5e110]	Qualification is a moat once paid, but it caps share. No data moat evidenced
4	Hardware-inclusive producers can return venture outcomes	SendCutSend profitable 8 years [src-20260519-38c7ed]; Hadrian plants are debt-funded [src-20260817-c48f28]	“Not … a fantastically profitable business”; a six-machine plant 1 cost “well north of $10 million” [src-20260817-8867f2]; net per-part saving after capex is 20-30% at best [src-20250712-e5e110]; Hadrian is “better suited to debt than VC” (ex-Anduril PI) [src-20250709-43486e]	Unproven. Underwrite the speed premium and the debt capacity, not cost leadership

1. The thesis in plain English

What it is. AI-native industrial production means companies that build and run their own factories (or, in one variant, the software that runs other people's factories) and sell the output: parts, metal, material. The product is the physical thing; the software is how they make it faster and with fewer skilled people than incumbents. SPC's slide 8 frames it as producers "that build and run their own production capacity, using in-house AI and software to quote, schedule, and control robots and machines" [src-20261022-80fd75].

Legacy vs AI-native.

	Legacy	AI-native
Who	Job shops (small owner-run machine or fab shops that build to a customer's drawing), service centers (metal distributors that stock, cut and ship metal), brokers and marketplaces	Hadrian, SendCutSend, Senra, 1872, Nox, Mariana
Software	ERP and CAM bought off the shelf (CAM = software that turns a 3D model into machine toolpaths; ERP = the back-office system of record)	Built in-house around quoting, nesting (packing parts onto a sheet or block to minimize scrap), CAM automation, scheduling, robot control and inspection
Speed	Average aerospace part 8-12 weeks, 40% late or with quality issues [src-20251024-2e5383] (med, single memo)	**Claim (SendCutSend):** 48 hours design to delivery [src-20260227-e92d45]; **Claim (Senra):** harness lead time six months to two weeks [src-20260128-a16c79]
Learning	Knowledge sits in the heads of machinists averaging early-60s in age (**Claim, Chris Power**) [src-20260908-65fab6]	Thesis: every part made improves the next quote, schedule and QC. Evidenced at 2 of ~15 names (section 6)

Three business models (taxonomy `model`):

Model	Best example	Tradeoff
owner-operator	SendCutSend: owns all its lasers and plants, bootstrapped to a $1.01B round in May 2026 [src-20260519-38c7ed], then a Series B at $2.1B five months later [src-20261020-487a39] (med, single trade write-up)	Owns the data and the margin; carries the capex and utilization risk
software-first	Arda: CAD-to-work-instructions agent inside Anduril's ArsenalOS; $160M Series A at $1.5B [src-20261020-914a64]	Capital-light, fast to revenue; needs a host plant for data and competes with Siemens and Rockwell on the installed base [src-20250326-cc1810][src-20260804-0e8130]
hybrid	Hadrian (owns factories, now embeds cells inside Lockheed's Grand Prairie plant on a 7-year deal) [src-20261014-015327]; Isembard (owned plus franchised factories on MasonOS) [src-20260909-b547c7]	Tries to get owner data with partner capital; harder to underwrite because economics are split

The core tradeoff in one line: owner-operators buy data ownership and pricing power with capital intensity; software-first buys speed and margin with data dependence on someone else's plant [inference].

1A. The big picture: macro primer for a newcomer

This section assumes no background. It explains what is happening in US and allied manufacturing, why money is flowing in now, how buyers actually purchase parts, and how venture investors think about it. Every number has a source. Where a number is my reading, it says [inference].

1A.1 Manufacturing in one page

- **How big it is.** US manufacturing added $2.90 trillion of value in 2025. That is 10.1% of GDP (gross domestic product, the total value of what the economy produces). It employs 12.65 million people, at average pay of $112,261 [src-20261013-a7c995].
- **How it is organized.** Think of a pyramid. At the top are a few **primes**: the big companies that sell finished systems (a missile, a jet, a submarine) to the government. The top five US defense contractors in FY2023 (fiscal year) were Lockheed Martin ($61.4B of contracts), RTX/Raytheon ($24.1B), General Dynamics ($22.9B), Boeing ($20.1B) and Northrop Grumman ($16.3B) [src-20251215-98f650]. Below them are **tier-1 suppliers** (big sub-systems), then **tier-2 and tier-3 suppliers** (individual parts, castings, forgings, wire harnesses, raw metal).
- **The bottom of the pyramid is small and old.** The US has 17,156 machine shops. 83% have fewer than 20 employees [src-20250626-23f6d9]. They are called **job shops**: small owner-run businesses that make parts to a customer's drawing. The count is shrinking: 4,874 machine shops were lost from 2002 to 2018, about 20% [src-20250909-29c826]. **Claim (Isembard):** the average owner is over 65 and 40% plan to retire within five years [src-20260310-45bab3].
- **The top has consolidated.** Aerospace and defense primes fell from 51 to 5 since the early 1990s [src-20220215-45c353]. Fewer buyers means fewer, larger programs, and more power over suppliers.

What this means. The supply base that makes physical parts is fragmented, aging and shrinking, while the buyers at the top are few and powerful. That gap is the opening every company in this primer is trying to fill [inference].

1A.2 Why this is happening now: six forces

1. Defense needs more munitions and ships, fast. After Russia invaded Ukraine in 2022, the US and allies used up munitions faster than factories could replace them. Congress's research service says much of the concern about the defense industrial base "has focused on munitions production due in large part to high rates of expenditure" [src-20251215-98f650]. Primes are now signing long deals with new producers. Lockheed signed a 7-year deal to put Hadrian's automated machining cells inside its missile plant [src-20261014-015327]. The Navy put $900M toward Hadrian plants for submarine parts [src-20260320-67ba4c]. - *What this means:* demand is strongest for defense parts that are late today. It depends on budgets that Congress sets each year [inference].

2. China controls key inputs. China leads refining for 19 of 20 strategic energy minerals, with about 70% average share [src-20250521-c62d29]. It makes about 94% of rare-earth magnets (the strong magnets inside motors, drones and missiles) and cut magnet exports in April 2025 [src-20251103-ee9b27]. The US imports 100% of 16 mineral commodities, 67% of rare earths and 57% of refined copper [src-20260206-04fe3e]. Almost 90% of the most advanced chips are made in Taiwan [src-20230928-c47558]. - *What this means:* "Can we make this at home?" is now a national-security question, not only a cost question. That is why governments pay for domestic capacity even when it costs more [inference].

3. Tariffs make imported metal and parts dearer. Tariffs are taxes on imports. Since April 2026, products made almost entirely of steel, aluminum or copper face a 50% tariff; products that contain them face 25%; and 10% if the metal was melted or smelted in the US [src-20260402-1b5d6c]. This

helps US mills: Nucor's gross margin rose from 14% to 20% (part of it a one-off refund) [src-20260812-77d6f5]. It hurts US companies that buy metal: Nucor's own fabrication segment reported "margin compression resulting from higher steel input costs" [src-20260812-77d6f5]. - *What this means:* tariffs are set by presidential proclamation and can be reversed. They are a tailwind, not a foundation [inference].

4. There are not enough skilled workers. Deloitte and The Manufacturing Institute project 3.8 million US manufacturing jobs to fill from 2024 to 2033, with 1.9 million possibly unfilled [src-20240416-b69154]. Manufacturing had 522,000 open jobs in August 2026, against a pre-pandemic average of 432,000 [src-20261013-a7c995]. The Defense Department says it needs at least 122,000 more shipbuilders, engineers and other critical workers by 2028 [src-20250101-af28c1]. - *What this means:* the bottleneck is people, not machines. Software that lets a less experienced worker do skilled work is the core promise of "AI-native" production [inference].

5. The government now lends and buys, not just regulates. Six tools matter (each explained in 1A.4): the NDAA, CHIPS, the IRA's 45X tax credit, DPA Title III, the Office of Strategic Capital and Section 232 tariffs. Together they lower the cost of building a plant and, more recently, promise to buy output for several years.

6. Money is available, but not cheap. Venture investors put large sums into this category in 2026 (Hadrian raised $1.37B in one round [src-20260807-b0dc5b]). But interest rates are rising again: the Federal Reserve raised its benchmark rate to 3.75-4.00% on September 16, 2026, saying "inflation remains elevated" [src-20260916-f0c016]. Factories are mostly paid for with loans once the first one works [src-20260817-c48f28]. Higher rates make every new plant more expensive to finance [inference].

What would reverse the trend. A defense budget cut or delay, tariffs struck down, a peace that slows munitions buying, or a jump in rates. Also: existing shops have spare capacity. The average machine in a US shop cuts metal only 34% of the time, against a 70% target [src-20250120-419768]. Some demand can be met without new factories.

1A.3 How a buyer actually buys a part (the procurement path)

This is the most important mechanism to understand. A new producer does not win "the market." It wins one part number at a time, at one buyer at a time.

1. **Request for quote (RFQ).** A buyer at a prime sends a drawing and asks suppliers for a price and delivery date.
2. **Supplier qualification.** Before a new supplier can make flight or weapon parts, the buyer checks its quality system. The common standard is **AS9100**, the aerospace quality management standard. Special processes (heat treating, coatings, inspection) need **NADCAP** accreditation. Buyers told us qualification takes 9-12 months at Raytheon [src-20250708-2aa3ad] and 1-2 years at Northrop [src-20260802-36d46c].
3. **First article inspection (FAI).** The supplier makes the first part. Every dimension, material and process is checked against the drawing and written up in a standard report (AS9102). FAI is repeated whenever the design, material, process, tool or location changes, or after two years without production [src-20190801-e81d7b]. *This is why moving a part to a new supplier is slow and costly.*
4. **Prototype and low-rate production (LRIP).** Small quantities, often under fast contracts such as **OTAs** (Other Transaction Agreements, which skip much of the normal federal acquisition rulebook and can roll into production without a new competition) [src-20241219-9f8fcc].
5. **Rate production.** Steady, high volume. Here price per part and consistent quality decide who wins. A Boeing executive says at around 1,000 parts a traditional shop is cheaper than Hadrian

[src-20260908-ed94f0].

6. **Repeat orders and multiyear contracts.** The prize. Programs can last about 50 years, and suppliers for a 2035 aircraft are being chosen now [src-20260818-9b2d5c]. The FY26 NDAA (National Defense Authorization Act, the yearly defense policy law) now allows multiyear buying for SM-6, Tomahawk, AMRAAM, JASSM, LRASM and PAC-3 missiles [src-20251218-f0d6ef].

Two rules that limit any one supplier. Buyers cap how much they depend on one shop (SpaceX caps a shop at about 1/3 of its revenue) [src-20250708-342eaf]. After a factory fire at a major supplier, Boeing says it “will never” put one part family with one supplier again [src-20260818-9b2d5c].

What this means. Early wins are fast prototype work that buyers pay a premium for. The test of a real business is whether those wins turn into repeat rate-production orders. That is exactly Lilly’s question 1, and the evidence so far is thin (section 8) [inference].

1A.4 The policy toolkit, in plain English

Tool	What it is	What it does for producers	Source
NDAA	The yearly law that sets defense policy (not the money itself; money comes in separate appropriations bills)	FY26 version: multiyear munitions buying (Sec. 804(b)); an Industrial Base Fund for castings, forgings and machine tools, with purchase commitments of up to 10 years and advance payments (Sec. 867); a review of which spare parts advanced manufacturing can make (Sec. 1842)	[src-20251218-f0d6ef]
DPA Title III	Defense Production Act of 1950. Title III lets the government pay to expand domestic capacity: direct purchases, purchase commitments, even installing equipment in private plants	23 awards, $820.5M since Jan 2025, mostly rare earths and rocket motors; none yet to the startups we track	[src-20251215-98f650][src-20261019-68fd3d]
OSC	Office of Strategic Capital, a Defense Department lender created in 2022. It makes loans and loan guarantees for 34 critical technology areas, including advanced manufacturing	$620M loan to Vulcan Elements for a magnet plant; FY2026 funding of $97.8M can back up to $4.4B of loans, and the 2025 reconciliation law adds credit subsidy for up to $100B each for critical minerals and covered technologies	[src-20251103-ee9b27][src-20260908-f6558e]
CHIPS Act	2022 law: $52.7B for FY2022-2027 for US chip factories, research and workforce	Mostly helps chip fabs; it co-funded the Vulcan magnet deal with a $50M incentive	[src-20230928-c47558][src-20251103-ee9b27]
IRA 45X	A per-unit tax credit from the 2022 Inflation Reduction Act for US-made energy components. Critical minerals earn 10% of production costs	The 2025 tax law (P.L. 119-21) phases the minerals credit out 2031-2034 and bars Chinese-linked supply chains	[src-20251001-45a0c1]
Section 232	A 1962 law letting the President set tariffs on national-security grounds	50/25/10% metals tariffs (see 1A.2)	[src-20260402-1b5d6c]

Tool	What it is	What it does for producers	Source
Buy American / Specialty Metals	Laws requiring the Defense Department to buy US-made goods; specialty metals (titanium, nickel, some steels) must be melted in the US	Protect domestic demand for metal and parts	[src-20251215-98f650]

What this means. Policy is strongest for metals and minerals (loans, credits, tariffs). For machined parts, policy works mainly through prime contracts and the new multiyear authorities [inference].

1A.5 How big is the market?

No free source gives a clean market size (TAM, total addressable market) for "AI-native production." What we have: - All US manufacturing: $2.90T value added [src-20261013-a7c995]. - Metal service centers (distributors that stock, cut and ship metal): about $200B of revenue (med) [src-20251201-4b732d]. - Precision manufacturing: about $71B by 2026, machine-shop segment over $60B (med, one analyst memo) [src-20251024-2e5383]. - Global industrial robot installs: $18.6B in 2025, a proxy for automation spending [src-20260924-28b59b].

Section 3 has the full table. *What this means:* the markets are large, but each startup sells into a narrow slice (one part type, one buyer group). Share per buyer is capped (1A.3), so growth needs many buyers [inference].

1A.6 How venture investors see it

- **It costs a lot to build.** A first six-machine factory cost "well north of $10 million" [src-20260817-8867f2]. Hadrian's Mesa plant was about $200M [src-20250820-cfbf02]. Software companies do not need this.
- **Margins are lower than software.** Gross margin is revenue minus the direct cost of making the product, as a share of revenue. Software companies often run far higher than manufacturers. Benchmarks here: Proto Labs 44.5% [src-20260206-8bb619]; Xometry's marketplace 34.7% [src-20260224-a7b149]; metal service centers 18-32% [src-20260729-407fd7][src-20261021-16bb64]; a manufacturing roll-up "effectively … 30%" (claim) [src-20241001-28c2aa].
- **Who pays for what.** Venture equity pays for the team, the software and the first plant. Banks, equipment lenders and the government pay for plants two and beyond, once a plant is proven. Hadrian: "equity never funds factory construction" (claim) [src-20260807-b0dc5b]; it has a $360M bank credit line [src-20260817-c48f28].
- **The bull case.** If software lets one plant run at much higher utilization with fewer skilled workers, it can earn better margins than a job shop and win share. **Claim (Chris Power, Hadrian CEO):** utilization of 65-80% versus about 10% at primes [src-20260610-779f8b].
- **The bear case, with equal weight.** Several buyers say they pay the same or more, not less (section 0). Some see these firms as "better suited to debt than VC" [src-20250709-43486e]. Recent failures: Daedalus (an AI-run machining factory) went insolvent in August 2026 [src-20260801-4532f6]; Markforged went from a $2.1B valuation to a $42.5M sale [src-20261023-63156a]; Bright Machines' SPAC (a listing via a shell company) fell through [src-20240527-6a4884].
- **Exits are unproven.** The repo has no acquisition of an AI-native producer above $100M (Appendix B). Likely exit paths are sale to a prime or incumbent supplier, an IPO (stock listing) at scale, or continued private ownership with debt [inference].

What this means for SPC. The question is not "is manufacturing coming back?" It is "can a software-run plant earn better margins than a job shop, and turn fast prototype wins into multi-year orders?" Everything after this section tests that [inference].

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2. Why now: the drivers in detail, ranked

Section 1A gave the plain-English version. This section gives the evidence for each driver, ranked by how much it actually moves revenue for the companies we track [inference on the ranking].

2.1 Defense demand and munitions/shipbuilding ramps (structural for 5-10 years; budget-cyclical at the margin)

- Mechanism: primes cannot add qualified capacity fast enough; they are now signing multi-year capacity deals with new producers.
- Evidence: Lockheed signed a 7-year agreement to put Hadrian automated cells inside its Missiles and Fire Control plant for PAC-3 MSE, THAAD, PrSM and GMLRS, commissioning early 2027 [src-20261014-015327]. Navy put $900M alongside >$1.5B of Hadrian capital for three maritime plants (Virginia/Columbia submarine parts) [src-20260320-67ba4c]. Divergent will build Tomahawk structures for RTX [src-20260601-976843] and is opening a Long Beach factory sized for 30,000 missile airframes or 60,000 warhead casings a year [src-20261023-e02543]. Army selected Vulcan Elements for magnets for SkyFoundry, a 1M-drones-a-year target [src-20260923-c53c99].
- **Claim (Chris Power):** submarine labor hours per boat rose from ~13M to 70M+; Virginia/Columbia built at ~0.6/yr vs ~2 bought; the enterprise is ~50M labor hours a year short [src-20260610-779f8b].
- Expert view: DoD supply base is “a limited pool of qualified suppliers operating at high utilization”; one Raytheon program manager reports 80-90% utilization at suppliers and 9-10 months for castings [src-20250708-2aa3ad].
- Layers: precision-parts, fabrication, metals-supply (magnets). Durability: high while munitions stockpiles are rebuilt [inference].

2.2 Skilled labor gap (structural)

- Headline numbers are in section 1A.2 (Deloitte 3.8M jobs, 522,000 openings, 122,000 defense workers). The 2026 Deloitte follow-up counts 2.3M technician openings 2025-2030 but excludes machinists, so do not add it to the 3.8M [src-20260910-a2cd42].
- Mechanism: the binding constraint is machinists, welders and programmers, not machines. Software that deskills tasks expands who can run a plant.
- **Claim (Hadrian):** Opus trains new operators in 30-40 days instead of a decade [src-20260926-70087f]; 40-hour tasks to 2 hours [src-20261022-002ed8].
- Layers: all producing layers; strongest in precision-parts and fabrication.

2.3 Reshoring and tariffs (policy-driven; reversible)

- Section 232 (a 1962 statute letting the President set tariffs on national-security grounds): steel and aluminum went from 25% to 50% in June 2025; the April 2, 2026 proclamation (effective April 6, 2026) sets 50% on articles made entirely or almost entirely of steel, aluminum or copper, 25% on derivatives, and 10% on derivatives made with US-melted or US-smelted metal; grid and industrial equipment gets a temporary 15% (10% with US inputs) until Jan 1, 2028 [src-20260402-1b5d6c].
- Effect visible in incumbents: Reliance cites aluminum prices “historically elevated largely due to” the 50% tariff [src-20260728-ae7dea]; Nucor cites Section 232 plus AD/CVD reducing imports, gross margin 20% vs 14% a year earlier, but the 20% includes a one-off $130M cash refund on prior-period raw-material costs (about 1.3 points), so not all of it is tariff-driven [src-20260812-77d6f5]. Tariffs squeeze fabricators: Nucor’s steel products segment (joists, deck, rebar fabrication) reported “margin compression resulting from higher steel input costs” [src-

20260812-77d6f5]. Tariffs help mills and hurt metal buyers, which matters for the fabrication layer [inference].
- Mechanism: a 10% vs 25% gap rewards anyone who can document US melt provenance, which favors software-native suppliers over brokers [inference, also argued in src-20260402-1b5d6c takeaways].
- Layers: metals-supply first, fabrication second.

2.4 Supply-chain trust and offshore concentration (structural)

- China's share of minerals, magnets and US import reliance: see section 1A.2.
- Offshore lead time: **Claim (SendCutSend):** China lead times two to three weeks vs its 48 hours, and US steel costs ~2x China [src-20260227-e92d45]; **Claim (Belosic):** can meet or beat China on only ~20-25% of work [src-20260519-2d6ceb].
- Layers: metals-supply (refining, magnets) most; precision-parts via ITAR (US-persons-only export control on defense articles) which keeps work onshore regardless.
- Bear: an ex-Anduril hardware lead says Taiwan is only 20-30% cheaper but more reliable, and Vietnam/Malaysia stay "significantly cheaper" even with tariffs [src-20250709-43486e]. Offshore cost gaps have not closed for volume work.

2.5 Capital availability (cyclical; currently abundant)

- Venture: Hadrian $1.37B Series D at $7.87B [src-20260807-b0dc5b]; Mariana $310M B [src-20260803-30e6d5]; Arda $232M total [src-20261020-914a64]; Mind Robotics >$1B in ~12 months [src-20260513-83aa83].
- Bank debt: Hadrian $360M revolver, Morgan Stanley lead plus 7 banks [src-20260817-c48f28]. **Claim (Mehr):** Machina equipment debt ~2-3% over prime [src-20260320-1361e2].
- Government: Office of Strategic Capital (OSC, DoD's lender for critical-tech companies) [src-20260908-f6558e]: $620M direct loan to Vulcan in a $1.4B package [src-20251103-ee9b27]. IEA: public minerals finance commitments quadrupled 2023-2025 to $65B [src-20260716-6a06e4].
- OTAs (Other Transaction Agreements, fast non-FAR prototype contracts that can roll into production without re-competition; $500M prototype cap) [src-20241219-9f8fcc]: DIU OTA to Freeform [src-20260901-6e197d].
- Read: equity funds plant #1; banks and government fund plants #2+ once a plant is proven [inference, consistent with src-20260807-b0dc5b].

2.6 Cost and capability of AI, vision and robotics (enabling, not sufficient)

- What changed in 24 months (from the repo): US robot installs back to growth, 38,428 in 2025 (+12%), US passed Japan for #2 [src-20260924-28b59b]; global installed stock 5.08M [src-20260924-28b59b]. Path Robotics' Obsidian weld model is now a supplier to 1872 [src-20260722-3bc2d8]; Freeform runs H200 clusters on site for real-time meltpool control [src-20260219-2125d7]; LLM agents now read CAD and write work instructions (Arda: 10 hours to 15 minutes at Anduril) [src-20261020-914a64].
- Counterpoint: US metalworking robot adoption is small, ~3,000 units a year [src-20260618-c903cc]. Hadrian's former head of programming says Opus relies on mature third-party CAM for toolpathing and its edge is workflow, not unique technology [src-20261004-9b47f5].
- Gap: the repo has no source quantifying the cost decline of vision sensors, GPUs or robot cells over 24 months.
- Layers: factory-ai and materials-rd depend on this most.

What would break “why now”

Tariff reversal or court limits would hit metals-supply and fabrication margins first (the 10%/25%/50% structure is a proclamation, not a statute) [inference]. A flat defense budget or delayed munitions buys would stall precision-parts plants built ahead of contracts; Power himself says ~$1B of capex must be deployed ahead of contracts [src-20260921-876080]. Higher rates matter because plants #2+ are debt-funded [inference]. AI over-claiming is already visible: the data loop is evidenced at only 2 of ~15 names (section 6), and Mehr publicly criticizes “SpaceX cosplay and an AI bumper sticker” [src-20261021-da5fb5]. Robot learning is far from production reliability: one-shot task success ~59-66% vs 99.9%+ needed [src-20260904-267293]; an investor notes even 99% accuracy is too low on a line [src-20261013-c6e3c0]. Demand is not the only constraint: average VMC in-cut spindle utilization is 34% vs a 70% target, so existing shops have slack before anyone needs new plants [src-20250120-419768] (med, survey). The domestic base has shrunk for 20 years (4,874 machine shops lost 2002-2018, -20%) and defense rides on commercial tier 2-3 volume, which VC-built defense plants do not replace [src-20250909-29c826].

3. Market size by layer
In one line: the markets are large, but there is no clean free estimate of the slice these startups can win.
No layer has a clean, free TAM. Where only a proxy exists, it says so.

| Layer | Figure | Year | Source | Conf. | Note | |—|—|—|—|—|—| | All manufacturing | $2.90T value added; 12.65M employees; $112,261 avg comp | 2025 | NAM facts [src-20261013-a7c995] | high | 10.1% of GDP | | metals-supply (distribution) | ~$200B revenue, ~9,400 establishments, top 50 ~50% | 2025 | First Research [src-20251201-4b732d] | med | Service centers only, not refining | | metals-supply (refining) | Market size not found; public finance commitments $65B | 2025 | IEA [src-20260716-6a06e4] | high | Gap: US refining TAM | | precision-parts (shop base) | Machine shops (NAICS 332710): 17,156 establishments, 223,313 employees, $14.23B payroll; 83% under 20 staff; down from 17,433 in 2022 | 2023 | Census CBP [src-20250626-23f6d9] | high | Counts, not revenue. IBISWorld says 16,619 businesses [src-20260201-adef50] (med) | | fabrication (shop base) | Architectural and structural metals (3323): 14,533 est, 423,013 employees, $28.26B payroll; sheet metal work

4,652 est; structural fab 3,434 est | 2023 | Census CBP [src-20250626-23f6d9] | high | No free revenue TAM | | precision-parts | ~$71B precision manufacturing by 2026; machine-shop segment >$60B | 2026 | Contrary memo [src-20251024-2e5383] | med | Single analyst memo, citations not traced | | materials-rd | No TAM in repo | n/a | n/a | n/a | Gap | | factory-ai | Global robot installs value $18.6B | 2025 | IFR [src-20260924-28b59b] | high | Proxy for automation spend, not software |					
Fragmentation (supports a consolidation or software-native entrant story): - 83% of the 17,156 US machine-shop establishments had under 20 employees in 2023 [src-20250626-23f6d9] (high). Contrary says ~90% have <20 employees and <$7M revenue [src-20251024-2e5383] (med). - DoD: aerospace/defense primes fell from 51 to 5 since the 1990s; castings and forgings face high barriers to entry [src-20220215-45c353].					

4. The five layers

In one line: the category splits into five layers, from raw metal at the bottom to factory software at the top. Each has different economics.

4.1 metals-supply (mining, refining, magnets, service centers)

- **Players:** Mariana Minerals (copper, lithium; owner-operator) [src-20260803-30e6d5]; Vulcan Elements (NdFeB magnets) [src-20251103-ee9b27]; Nox Metals (automated service center, `passed`) [src-20261009-404b9b]; candidates Hertha (iron/steel), Cyclic, Mass Magnetics, Duranium (`_candidates.md`). Incumbents: Reliance, Ryerson, Nucor, MP Materials.
- **Economics:** service centers earn 18-32% gross margin (Ryerson 17.7% Q2’26 [src-20260729-407fd7]; Reliance 27.9% LIFO, 31.6% FIFO Q3’26 [src-20261021-16bb64]). Refining economics: gap.
- **AI-native change: Claim (Mariana):** MarianaOS halves 5-10 year project timelines [src-20260803-30e6d5]; mine ~50% autonomous with RL agents adjusting process control every 15 minutes [src-20260623-d1790b]. **Claim (Nox):** quote in minutes vs 3-5 days at brokers [src-20250813-5ee11b].
- **Risks:** commodity price; permitting; offtake concentration (Vulcan’s offtakes are mostly government) [src-20251103-ee9b27].
- **Why it matters:** policy pull is strongest here (OSC, Section 232, IEA) [inference].

4.2 fabrication (sheet metal, welding, forming, harnesses)

- **Players:** SendCutSend (owner-operator) [src-20260519-38c7ed]; Senra (wire harnesses) [src-20260701-027c1b]; Machina Labs (robotic sheet forming, hybrid) [src-20260204-ce7053]; 1872 (robotic steel fabrication) [src-20260722-985690]; candidates Atomic Industries (tool and die) [src-20261023-a3bfe7], Advanced Metal Research.
- **Economics:** SendCutSend profitable for eight years [src-20260519-38c7ed]; ~$140M run rate per a deleted post (low) [src-20260504-c3ded5]; “nine digits” of revenue per Sequoia’s Reed [src-20261020-487a39] (med).
- **AI-native change:** instant quoting plus auto-nesting; Senra quotes in ~1 day and claims 99% first-pass yield (FPY: share of parts passing inspection first time, no rework) [src-20260701-027c1b].
- **Risks:** low-mix volume ceiling; Machina cycle times are 30 minutes to hours per part vs seconds for stamping [src-20261023-c3b62f].

4.3 precision-parts (CNC machining, additive, castings, composites)

- **Players:** Hadrian (hybrid) [src-20260807-b0dc5b]; Divergent (hybrid) [src-20250915-311606]; Isembard (hybrid franchise) [src-20260909-b547c7]; Freeform (owner-operator AM) [src-20260219-2125d7]; Amca (acquire-and-upgrade) [src-20260520-af3e6c]; Fabri (casting) [src-20260617-4dfb96]; Layup (composites) [src-20260602-a7b79e]; Hybron [src-20260413-abc797]. Incumbents: Howmet, PCC, ATI, Carpenter, Proto Labs, Xometry.
- **Economics:** Proto Labs FY25 GAAP gross margin 44.5% [src-20260206-8bb619], factory 49% vs network 31% [src-20260206-2b13c7]; Xometry marketplace 34.7% [src-20260224-a7b149]; Howmet engine products EBITDA margin 37.7% [src-20260806-9a83f7]. **Claim (Rolls-Royce engineer):** small shops want low double-digit margins [src-20251120-0c271b]. **Claim (ex-Hadrian programming head):** a typical shop can go from 15% to 30% margin in 12-18 months with Hadrian-style ops [src-20261004-9b47f5].
- **AI-native change: Claim (Raytheon manager):** Hadrian ~50% lower price at ~1/3 lead time [src-20250708-2aa3ad]. Contradicted below (section 8).

- **Risks:** customer qualification (AS9100 = aerospace quality management standard; NADCAP = industry accreditation for special processes like heat treat and NDT) takes 9-12 months for suppliers [src-20250708-2aa3ad] and 1-2 years per Northrop [src-20260802-36d46c]; layer is now priced at $7.87B for the leader [inference: priced out for SPC]. **Failures in this exact model:** Daedalus (ex-OpenAI founder, AI-run CNC factory, Karlsruhe, $21M A led by NGP) entered insolvency Aug 2026 [src-20260801-4532f6][src-20251203-fc28ea]; VulcanForms replaced its founder CEO [src-20240919-72efcc], then raised $220M [src-20260130-365541]. Ex-Hadrian engineer: a six-machine first factory cost "well north of $10 million" and "manufacturing is not actually a fantastically profitable business" [src-20260817-8867f2].

4.4 materials-rd (self-driving labs, AI materials discovery)

- **Players:** Periodic Labs [src-20261022-682e62]; CuspAI (software-first) [src-20261022-37a0ea]; Radical AI (owner-operator lab) [src-20260617-f9d293]; Lila Sciences (candidate). Incumbent-adjacent: Microsoft Discovery, Citrine.
- **Economics: Claim (CuspAI primer, AlphaSights):** custom programs 5-15% gross margin, platform 85-90%; pilot conversion 40-50% [src-20261016-29997e] (med).
- **AI-native change: Claim (Radical):** 8-20 alloys a day, target 100/day; $60-300 per experiment; samples grams to 500 g, no manufacturing yet [src-20260617-f9d293]. **Claim (Fedus, Periodic):** at full throughput, 4 months of experiments in under 2 weeks [src-20260917-1bcb81].
- **Risks:** no revenue disclosed at any of the three; valuations ($2.6B to $7.5B) precede product [inference]. Track record of the underlying science is mixed: Berkeley A-Lab's claimed 41 new materials were challenged as mostly misidentified known phases by an outside critique that found "no new materials were discovered" [src-20240124-513fd9]; the claim was challenged, not retracted; MatterGen is limited to ~20 atoms per cell and needs DFT validation [src-20261022-3e2a9d]; Lila's catalysts are unproven in a real electrolyzer [src-20261019-392916]. Business model is unproven: an Albert Invent director says chemical majors have not bought royalty models and the problem is "millions of SKUs", not crystal structures [src-20260913-3ad55e].
- **Buyer view (expert calls, read line by line).** Former Moderna director: buyers reach these platforms through CROs and pay per successful candidate; of the last 10 campaigns CuspAI won 8, Radical 1 and Lila 1 (claim); long-run potential ranked Radical > Lila > CuspAI > Periodic; CRO book-to-bill was "very bad" in 2024-25 [src-20260909-af3e60] (low: several claims cannot be checked). Albert Invent director (a competitor, so biased): ACVs run from hundreds of thousands to millions; procurement "shut[s] down" big sticker prices; the proof that opens budgets is a product launched in "one-third the number of iterations"; his $100 split is CuspAI $50, Lila $35, Periodic $15 [src-20260913-3ad55e].

4.5 factory-ai (software for other people's plants)

- **Players:** Arda (CAD to work instructions) [src-20261020-914a64]; Mind Robotics (Rivian partner, robot foundation models; The Verge corrected "spinoff" to "partner of Rivian, but not a spinoff") [src-20261013-cc78c0]; candidates Rapta, Allus, Faraday. Incumbents: Siemens (with Altair, Dotmatics) [src-20250326-cc1810][src-20250701-a7fef9], Rockwell (GM 49.5%) [src-20260804-0e8130]. MES (manufacturing execution system: software that tracks and directs work on the shop floor) is the incumbent category.
- **Economics: Claim (Arda):** usage-based pricing [src-20261020-a1e174]; no revenue disclosed.
- **AI-native change:** work-instruction agents are the only shipped wedge (Arda at Anduril) [src-20261020-914a64] [inference].
- **Risks:** distribution vs Siemens/Rockwell; data rights sit with the host plant [inference]. Investor view: model capability is not the constraint; value needs unified plant data and workflow change [src-20261006-6265ad]. Mind Robotics has raised >$1B and still shown no robot publicly [src-

20261013-cc78c0]. Bright Machines, the earlier software-defined-factory bet, failed its SPAC and pivoted to AI-server assembly [src-20240926-fd90fb].

- **AI CAM and inspection (promoted from the skipped pile in this audit).** CloudNC's CAM Assist generates 3-axis and 3+2-axis machining strategies in minutes inside GibbsCAM, with the programmer reviewing each step; CloudNC runs its own Chelmsford factory as a test bed [src-20261021-9769d1][src-20261009-7cafd0]. AI Robotics (DGIST spinout) runs weld-bead inspection on 24 lines in 3 countries on more than 4M field data points [src-20261011-a8b14d]. Both are software-first wedges that give incumbents the same tools [inference].

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5. Company profiles

Every file in `companies/` whose status is not `not-a-fit`. Full profiles for tracking, passed and `candidate` files with a deep dive; one-line table for the remaining candidates and incumbents. Ordered by relevance to an SPC Series B-D check [inference on ordering].

5.1 Senra Systems | fabrication | owner-operator | tracking

- **What:** wire harnesses and cable assemblies for defense and space, own plants (~95,000 sq ft over two sites) [src-20260701-027c1b].
- **Capital:** $65M Series B Jul 2026 [src-20260715-68183e].
- **Customers:** Anduril, L3Harris, Northrop [src-20260128-a16c79]; growth mostly from existing customers and referrals [src-20260701-027c1b].
- **Software:** Amp, in-house platform touching everything but finance: AI BOM read, live distributor pricing, auto labor estimates, work instructions [src-20260701-027c1b].
- **Claims (Senra):** 99% FPY [src-20260701-027c1b]; 1,000 harnesses/month going to 10K/month in 2027 [src-20260715-68183e]; sweet spot 100-1,000 units a run, quotes in under 2 weeks [src-20260701-027c1b]; by its CEO's own account automation is still early [src-20260715-821445]; ~190 staff to ~300 by YE2026 [src-20260701-027c1b].
- **Fit:** owns-production, in-house-software, mission-critical-output, customer-qualification. Data-loop partial. [inference]
- **Why it matters:** best-fit Series B-sized owner-operator in the repo; harnesses are ~95% outsourced even at SpaceX (**Claim, Senra**) [src-20260715-821445].
- **Bear:** every operating metric (99% FPY, 10K/month, growth) is company-stated; no customer or expert call in the repo; only 10 sources mention Senra. Harnesses are labor-heavy, so the software edge may be workflow rather than automation [inference].

5.2 SendCutSend | fabrication | owner-operator | tracking

- **What:** instant-quote sheet metal, CNC and finishing; owns all plants (Reno; Paris, KY; Arlington, TX) [src-20261020-487a39].
- **Capital:** $110M at $1.01B, first institutional round, May 2026 [src-20260519-38c7ed]; Series B at $2.1B led by Spark, Oct 2026, size undisclosed [src-20261020-487a39] (med, single trade write-up).
- **Scale:** 32K orders/month, ~100% y/y [src-20260525-4a12ab]; ~$140M run rate (deleted post) [src-20260504-c3ded5] (low); 650 staff [src-20260904-1929b3] vs 169 [Harmonic].
- **Claims (Belosic):** 48-hour turnaround [src-20260227-e92d45]; 80-90% utilization target [src-20260626-541ac8]; new facility ~$50M funded with equipment loans [src-20260519-2d6ceb]; "I love taking VC dollars and putting it directly in the payroll" [src-20261020-487a39].
- **Fit:** owns-production, in-house-software, speed-moat. Mission-critical-output weak (60% of Fortune 500 but commercial mix) [src-20261020-487a39].

- **Why it matters:** proof that a software-native owner-operator can be profitable without defense budgets [inference]; now likely priced beyond SPC.

5.3 Hadrian | precision-parts | hybrid | tracking

- **What:** automated CNC factories run on Opus software; now also embedded cells at Lockheed [src-20261014-015327].
- **Capital:** Series C $260M equity+debt Jul 2025 [src-20250721-d99a2a]; ~$1.6B valuation Jan 2026 [src-20260807-b0dc5b]; $1.37B Series D at $7.87B, Aug 6 2026 [src-20260807-b0dc5b]; $360M revolver [src-20260817-c48f28]. Total ~$2.24B [Harmonic].
- **Footprint:** four facilities (CA, AZ, AL), about 2.85M sq ft; 24 machines inside Lockheed Grand Prairie plus 24 in Arizona for overflow, line live by January 2027 (**Claim, Power**) [src-20261015-1a282b].
- **Capex:** ~$200M Mesa factory, 270K sq ft, 350 jobs [src-20250820-cfbf02]; six factories since 2022, latest 2.2M sq ft [src-20260921-876080].
- **Experts:** see section 8; share of wallet 2-3% at Boeing [src-20260802-ef6506], <10% at Northrop [src-20260802-36d46c].
- **Fit:** all seven tests partly met; customer-qualification still building (**Claim, Power:** AS9100-grade trust takes years) [src-20260908-65fab6].
- **Bear (expert calls, med):** ex-Anduril hardware lead: Hadrian “struggled to consistently deliver high-quality, complex parts” and got no production volume there [src-20250709-43486e]; Boeing CTO-org manager: “we have actually paid probably more”, and no certified flying aircraft uses its parts [src-20260818-9b2d5c]; Northrop director: incumbents still have higher precision and a cost edge, and Ti/Inconel capability lags [src-20250707-3b7a46]; defense-startup chief of staff chose Atomic and questions how automated Hadrian really is [src-20250729-a0a78c]; ex-Xometry VP: deals are slow to win [src-20260808-dc5cbb]; ex-Hadrian engineer: CAM vendors are the main threat, and buyers cap a supplier at ~1/3 of its revenue [src-20250708-342eaf].
- **Buyer and insider voices (expert calls read line by line; speaker role as given by the call network; med):**

Speaker (relation)	What they said	Source
Boeing Defense head of product, CTO org (current customer)	“None of their parts right now are flying … all in demonstration stage.” Near term a gap filler for spot buys of about 100 parts; long term “about 50%-60% automation”. Scores: speed 8, process 9, scalability 8 out of 10. “We get very excited when we get a new supplier that gives us a discount of 5%-7%”	[src-20260818-9b2d5c]
Boeing strategy and M&A lead (buyer view)	Brackets, hinges and bulkheads in under 1 month vs 3-6 months; a “backup player” and “low-volume shop”; price about PCC or slightly more; automation cuts labor from 100 to 30-50 hours but capex offsets it	[src-20250712-e5e110]
Northrop operating-unit GM (buyer)	“Cost is secondary.” “Machining the part is just 1/3 of the battle.” Hadrian handles fasteners, shims and surge work; needed 8 parts, charged for 10-15; rates it 8/10	[src-20250707-3b7a46]
PI, ex-Anduril hardware (buyer)	Not “able to deliver reliably high quality parts”; a “narrative play”; Anduril’s CNC prototypes went mostly to Protolabs	[src-20250709-43486e]

Speaker (relation)	What they said	Source
Chief of staff, 40-person defense startup (buyer)	Uses Atomic Industries more (“turn production in eight days”). Hadrian also claims 8-9 days but is “more manual” and heavy on program managers	[src-20250729-a0a78c]
GE program lead (indirect exposure)	Hadrian is “too new, too young”; broker markups are “pretty horrific”; UK MOD DE&S has a new direct-sourcing cell	[src-20250709-0250ae] (low)
Ex-Hadrian head of engineering	Would be “surprised if any one company could capture 1%” of ITAR machining; machines are German and cost about $1M each; “only have 10 customers”; “all but one of the first 12 or 15 employees are gone”	[src-20250708-342eaf]
Ex-Hadrian (second call)	Power rejected a roll-up as “unredeemable” and went greenfield; started with aluminum milling only; “automate 80% of the tasks about 80% of the way”; keeps 10% of capacity idle for surge	[src-20260611-7b80f0]
Ex-Hadrian (third call; holds equity, a bias)	Factory 1: six machines, “well north of $10 million”; in years 1-2 Anduril was the only defense payer; CAM pay $130-150K vs $80-110K market; “dangerously good salesman”	[src-20260817-8867f2]
Ex-Xometry VP (peer)	“Hadrian needs 10 factories … Xometry wants 10,000 customers”; revenue “surprisingly small” vs investment; rates Hadrian 6/10	[src-20260808-dc5cbb]

- **Contradictions, side by side.** Lead time: Northrop says Hadrian is fast [src-20250707-3b7a46] | a defense startup finds Atomic as fast and less manual [src-20250729-a0a78c]. Quality: Boeing scores process 9/10 [src-20260818-9b2d5c] | ex-Anduril PI says parts missed spec [src-20250709-43486e]. Competitor Formlogic: named as a live competitor by an ex-Hadrian engineer [src-20250708-342eaf] | “went under” per the client on another call [src-20250709-43486e] (unverified).
- **Why it matters:** the category benchmark; at $7.87B a reference not a target [inference].

5.4 Isembard | precision-parts | hybrid | tracking

- **What:** owned plus franchised machining factories on MasonOS; US, UK, DE, FR, UA [src-20260310-45bab3].
- **Capital:** $9M seed plus $50M Series A (USV lead), Mar 2026; valuation undisclosed [src-20260310-45bab3]; Harmonic estimate ~$250M [Harmonic] (low).
- **Claims (Isembard):** 40-60% throughput gains, 30% lower opex [src-20250929-857f17]; franchisee live in under 90 days [src-20260909-b547c7]; turning down orders [src-20260810-f0371c].
- **Customer view:** a humanoid company used Isembard for two urgent $2.3-6K orders, 2-day turnaround [src-20260621-f96bbd]; Airbus supply-chain head sees networks as prototype suppliers [src-20251106-dfddcb].
- **Bear:** Thales Alenia Space CPO is “skeptical” for aerospace, defense and space: he must qualify and see each production site, which a franchise network obscures; not a customer [src-20251110-84b67e]. Throughput and opex claims are company-only. BBC: ~100 staff at the 160,000 sq ft Southwark site [src-20261019-202dd7].
- **Fit:** franchise variant of Hadrian at a fraction of the price; capital-light [inference].

5.5 Machina Labs | fabrication | hybrid | tracking

- **What:** RoboCraftsman robotic sheet forming plus machining/welding; owns plants and places cells [src-20260204-ce7053].
- **Capital:** $124M Series C Feb 2026 [src-20260204-ce7053].
- **Customers:** biggest is military depots (legacy panels whose dies no longer exist) [src-20261022-ca4c7a]; ~70% defense; JASSM qualification; CMMC L2 [src-20260204-ce7053]; 5 Air Force awards, $7.8M, largest $3.37M for incremental sheet forming (Feb 2025), plus a $507K Robins AFB contract to maintain an installed Machina cell; Air Force is a repeat buyer [src-20250416-317e81]. A Rolls-Royce-side expert says Machina is prototype and tooling-less forming, “not going to involve in our serial production” [src-20250623-5bc67c] (med).
- **Claims (Mehr):** up to ~50x faster lead times [src-20261022-ca4c7a]; staff 70 to 240 [src-20260320-1361e2].
- **Claim (Machina):** new 200,000 sq ft plant, up to 50 RoboCraftsman cells, “thousands” of assemblies a year [src-20260204-ce7053].
- **Risk:** cycle time per part 30 minutes to hours vs seconds for stamping [src-20261023-c3b62f].

5.6 Freeform | precision-parts | owner-operator | tracking

- **What:** AI-controlled multi-laser metal AM service bureau; does not sell printers [src-20241024-7f1c2e].
- **Capital:** $67M Series B Feb 2026; PitchBook $179M post [src-20260219-2125d7] (med); ~$126M raised.
- **Skyfall: Claim (Freeform):** laser platform “scheduled to go live in the first half of 2026”, 25x capacity, thousands of kg of parts a day [src-20260220-348ad9]. No source confirms it went live.
- **Loop:** microsecond meltpool vision, FPGA/GPU processing, models adjust print parameters live [src-20241024-7f1c2e]; on-site H200 clusters [src-20260219-2125d7].
- **Demand:** hundreds of mission-critical parts weekly, customers undisclosed [src-20260219-2125d7]; DIU OTA [src-20260901-6e197d].
- **Bear:** metal AM has the worst failure record in the theme (Velo3D, Markforged, Desktop Metal-era SPACs; section 7); Freeform’s loop and capacity claims are company-only, customers undisclosed, Skyfall unconfirmed.
- **Why it matters:** clearest closed-loop process control in precision-parts; small enough for SPC [inference].

5.7 Divergent | precision-parts | hybrid | tracking

- **What:** AM plus robotic assembly (DAPS). Named defense work: Raytheon Tomahawk midbody second source and CoAspire munitions [src-20260601-976843]; a teaming agreement (not a production award) to explore DAPS for L3Harris Red Wolf/Green Wolf [src-20261013-eda149]; General Atomics work is a 2023 partnership on small drone (UAS) aerostructures, not missiles [src-20261022-8629b9]. **Claim (Czinger):** some programs about 50% cheaper per unit [src-20260523-b05ded] (unverified). An Anduril engineer called Divergent “not really a scalable thing” (Mar 2025, opinion) [src-20250623-5bc67c] (low).
- **Capital:** Series E $290M, of which $40M debt, Sep 2025 [src-20250915-311606].
- **Scale:** Long Beach plant, 430,000 sq ft, 1,000 jobs, online by YE2026, 30,000 airframes a year [src-20261023-e02543].
- **Claims:** revenue up >5x [src-20250915-262bfd]; ~30 programs with 15 primes [src-20261014-47f64b]; A&D ~80% [src-20261023-e02543]; Tomahawk casting yields under 20% at legacy suppliers (claim) [src-20260113-c08796].
- **Why it matters:** the auto-to-defense pivot that worked; benchmark at $2.3B [inference].

5.8 Amca | precision-parts | owner-operator | candidate

- **What:** buys AS9100 plants (8) and re-engineers components with the RAPID AI platform [src-20260520-af3e6c][src-20261008-149196].
- **Capital:** $300M Series B at $1B+, Caffeinated and Lightspeed [src-20260520-af3e6c].
- **Demand:** first Army Advanced Manufacturing CSO Active Capability Gap FFP award, value undisclosed [src-20261008-149196]. **Claim:** qualifies re-engineered parts in under 90 days [src-20261008-149196].
- **Why it matters:** the roll-up path Hadrian rejected [src-20261004-9b47f5]; buys qualification instead of earning it [inference].

5.9 Vulcan Elements | metals-supply | owner-operator | candidate

- **What:** sintered NdFeB magnets; 10 t pilot in Durham, 10,000 t plant planned [src-20251103-ee9b27].
- **Capital:** $65M Series A (Altimeter) [src-20261010-8e6ec0]; $1.4B package: $620M OSC loan, $50M Commerce equity, $550M private [src-20251103-ee9b27]; Form D $430.7M sold [src-20260402-c8e4dc].
- **Demand:** 7 federal awards, $5.7M incl. an Army magnet IDIQ [src-20260710-6361ea][src-20250501-1885f8]; SkyFoundry magnets [src-20260923-c53c99]. **Claim (Maslin):** pilot sold out 12 months ahead; in-house characterization data feeds ML recipe models [src-20250811-e7cb9a].

5.10 Mariana Minerals | metals-supply | owner-operator | tracking

- **What:** builds and runs mines and refineries (Copper One, Lithium One) on MarianaOS [src-20260803-30e6d5].
- **Capital:** $310M Series B (Khosla), ~$400M parent plus project capital; valuation undisclosed in release [src-20260803-30e6d5]. $1.5B appears only in Harmonic and secondary coverage [Harmonic] (low).
- **Ops:** ~250 staff, ~20% software/ML; Utah mine ~50% autonomous [src-20260623-d1790b]. **Claim (Caldwell):** 30% lower refining cost, 40-50% lower mining cost [src-20260502-21f16a]; 10 projects in 10 years [src-20260513-dd56eb].
- **Why it matters:** one of two names with a credible data loop; Series B check size above SPC [inference].

5.11 1872 | fabrication | owner-operator | tracking

- **What:** AI-native steel fabrication (skids, frames, enclosures) for energy and utility, Cincinnati [src-20261022-fdbf29].
- **Capital:** $15M seed via O.H.I.O. Fund [src-20260722-985690].
- **Ops:** 18 staff, ~30,000 sq ft, producing since Aug 2026 [src-20261016-d61bc1]; Path Robotics weld cells [src-20260722-3bc2d8].
- **Claims (website, no baseline):** 99% FPY, 70% arc-on time, 85% weld cost reduction [src-20261022-fdbf29]. Full autonomy targeted 2027 [src-20260722-985690].
- **Why it matters:** cheapest entry into owner-operator fab; pre-revenue proof [inference].

5.12 Nox Metals | metals-supply | owner-operator | passed

- **What:** automated metals service center outside Detroit; Factos nesting [src-20261009-404b9b].
- **Capital:** $11.5M seed (Hyperion) [src-20260616-1cf7cb].
- **Status:** passed; kept as a seed-stage reference for a software-led service center.

5.13 Arda | factory-ai | software-first | tracking

- **What:** agent reads CAD and design docs, writes assembly order and work instructions [src-20261020-914a64].
- **Capital:** $72M seed (Accel, Founders Fund); $160M Series A at $1.5B (Altimeter, Accel, Khosla); $232M total [src-20261020-914a64].
- **Claim (Arda/Anduril):** work instructions 10 hours to 15 minutes; complex engineering 30 hours to ~2.5 [src-20261020-914a64].

5.14 Mind Robotics | factory-ai | software-first | tracking

- **What:** trains robot models on factory work (pick/place, fastening, connectors) and sells robots, not parts [src-20261022-eede0b]. Rivian is partner, first customer and shareholder, not parent: “a partner of Rivian, but not a spinoff” [src-20261013-cc78c0]; customer concentration on Rivian [src-20260515-176e26].
- **Capital:** >$1B raised [src-20260515-176e26]. No public product.

5.15 Periodic Labs | materials-rd | owner-operator | tracking

- **Capital:** $300M seed, Sep 2025 [src-20251022-8eb7c0]; reported in talks for a much larger round (med) [src-20260514-5796ab]. No revenue disclosed.
- **Offer:** custom AI agents for clients’ engineers; one unnamed semiconductor client [src-20261022-682e62].

5.16 CuspAI | materials-rd | software-first | tracking

- **Capital:** $450M Series B, Kleiner and NEA [src-20260720-840ae3].
- **Network:** ~30 industry, ~15 lab partners incl. Samsung, AMD, Lam, Applied [src-20261022-37a0ea].

5.17 Radical AI | materials-rd | owner-operator | tracking

- **Capital:** $55M Seed+ led by RTX Ventures, a prime’s venture arm [src-20250718-d46f24].
- **Ops:** see 4.4; AF award $1.2M [src-20250523-f41dd1].

5.18 Terrafirma | adjacent | owner-operator | tracking

- $100M Series A, autonomous earthworks, Austin [src-20260714-976329]; ~$500M estimated post [Harmonic] (low). Adjacent to thesis.

5.19 Remaining candidate and incumbent files (one line each)

Company	Layer	Status	One line
Fabri	precision-parts	candidate	Automated investment casting, $13.5M, RTX and LMT Ventures; shipping castings [src-20260617-4dfb96]
Layup Parts	precision-parts	candidate	Custom composite parts, $42M A [src-20260602-a7b79e]
Hybron	precision-parts	candidate	Composite shells, $25M seed; contract floor >$100M [src-20260413-abc797]

Company	Layer	Status	One line
Atomic Industries	fabrication	candidate	AI tool and die, $25M A [src-20261023-a3bfe7]
Rangeview	precision-parts	candidate	Printed-mold castings, $4.2M seed [src-20260923-3ab808]
Furientis	adjacent	candidate	Low-cost interceptors, $25M seed (Benchmark), DoW OTA [src-20261006-410f77]
Firestorm Labs	adjacent	candidate	SBIR to Phase III IDIQ with repeat SOCPAC task orders, 19 awards, $53.8M [src-20251022-629fc0][src-20251219-bd5788]
Re:Build	precision-parts	candidate	Engineering plus manufacturing roll-up, 12 sites [src-20261006-1212f9]
VulcanForms	precision-parts	candidate	LPBF AM plus machining; CEO change then re-raise (pivot reference) [src-20240919-72efcc]
Proto Labs, Xometry, Howmet, PCC, ATI, Carpenter, Reliance, Ryerson, Nucor, Rockwell, Siemens	various	incumbent	Benchmarks in section 7

6. Data loop scorecard

In one line: the “data loop” is the idea that every part a factory makes teaches its software to make the next one cheaper or better. This table scores how much proof each company has.

Scale: 0 asserted only; 1 mechanism described; 2 partial evidence; 3 quantified improvement over time. Nobody scores 3.

Company	Score	Mechanism	Best evidence	Missing
Freeform	2	Meltpool vision adjusts each layer live	[src-20241024-7f1c2e] [src-20260219-2125d7]	Yield or cost trend over time
Mariana	2	RL agents adjust process every 15 min	[src-20260623-d1790b]	Recovery or cost data per tonne
Hadrian	2	Opus workflow, programming and QC	30-40 day training [src-20260926-70087f]; 25-75% cycle-time cuts at a studied shop; 50% cut in order-to-first-article within about two years [src-20261004-9b47f5]	Relies on third-party CAM [src-20261004-9b47f5]
Radical AI	2	Self-driving lab closed loop	8-20 alloys/day [src-20260617-f9d293]	Hit rate trend
Divergent	2	Generative design to print to assembly	600+ part numbers [src-20250915-311606]	Cost per part trend
Senra	1	Amp standardizes inputs, digital twin guides technicians, tracks materials and ECOs [src-20260715-68183e]	99% FPY claim [src-20260701-027c1b]	Before/after

Company	Score	Mechanism	Best evidence	Missing
SendCutSend	1	Instant quoting at volume	32K orders/month [src-20260525-4a12ab]	Any learning metric
Isembard	1	MasonOS ML quoting across franchise	Claim 40-60% throughput [src-20250929-857f17]	Third-party proof
Machina	1	Robot forming learns per geometry	Claim 50x lead time [src-20261022-ca4c7a]	Cycle-time trend
Vulcan	1	Characterization data into recipe ML	[src-20250811-e7cb9a]	Any number
Nox	1	Auto-nesting, quote-time estimates	[src-20260325-5b68db]	Scrap rate
Arda	1	Agent improves with host design data	[src-20261020-914a64]	Data rights
Periodic, CuspAI, Mind	1	Lab or factory as RL environment	[src-20260917-1bcb81] [src-20261022-37a0ea] [src-20260513-83aa83]	Production evidence
1872	0	Weld model is Path’s, not 1872’s	[src-20260722-3bc2d8]	Everything

Read: the loop is real where the process is physical and continuous (melt pools, ore, alloys), and weakest where the product is discrete parts sold on drawings [inference].

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7. Competition and incumbents

In one line: incumbents (established public companies) set the margin benchmarks, and several are adding AI themselves.

Incumbent	Metric	Value	Source	AI move
Proto Labs	FY25 revenue / GAAP GM	$533.1M / 44.5%	[src-20260206-8bb619]	AI pricing [src-20260731-ef3d40]
Proto Labs	Factory vs network GM	49% vs 31%	[src-20260206-2b13c7]	
Xometry	FY25 marketplace / total GM	34.7% / 39.1%; net loss $61.7M	[src-20260224-a7b149]	Siemens partnership [src-20260804-7625ed]
Xometry	Q2’26 revenue	$229M (+41%)	[src-20260804-33fe9f]	
Reliance	Q3’26 GM	27.9% LIFO / 31.6% FIFO	[src-20261021-16bb64]	
Ryerson	Q2’26 GM	17.7%	[src-20260729-407fd7]	
Nucor	Q2’26 GM	20% vs 14% (incl. one-off $130M refund, ~1.3 pts)	[src-20260812-77d6f5]	
Howmet	Q2 revenue / Engine EBITDA margin	$2.55B / 37.7%	[src-20260806-9a83f7]	AI process control [src-20260301-357ec4]
ATI	Adj. EBITDA margin / backlog	22.6% / $4.4B	[src-20260806-70254e] [src-20260628-27be55]	
Carpenter	GM	32.1% (40.4% ex-surcharge)	[src-20261022-d47483]	
Proto Labs	10-K AI uses	pricing, molding inspection, CNC toolpath checks, AI routing in minutes	[src-20260220-fe6c5e]	yes

Incumbent	Metric	Value	Source	AI move
Rockwell	GM	49.5%	[src-20260804-0e8130]	

What the expert calls add on incumbents (claims, med).

Incumbent	Data point	Source
PCC	Minimum account size about $50M+, majors >$25M; share of Boeing machining fell from 50% to 30% over 10 years; parts cost 5-7% more; also automating	[src-20250712-e5e110]
PCC	A factory fire hit all allocations from that plant, so Boeing will not concentrate supply again	[src-20260818-9b2d5c]
Boeing in-house (conflict)	About 40% of parts made in-house; flight-critical work stays in-house [src-20250712-e5e110] | only 20-30% of the 1.2M parts per airplane made in-house [src-20260818-9b2d5c]. Different scopes (spend vs part count) are possible	both
Boeing supply base	More than 10 machining shops, wants 2-3; legacy suppliers have “extremely high inertia”	[src-20260818-9b2d5c]
Xometry, Protolabs	Target $5M+ accounts; Boeing spend with each is under $1M and over $1M	[src-20250712-e5e110]
Xometry	Grew after shedding its own machines; Protolabs was held back by its own factories; market stays fragmented for 5 years	[src-20260808-dc5cbb]
Protolabs	8-inch cube limit on fast CNC; Xometry supplier rates “abysmal”	[src-20260611-7b80f0]

Read: the incumbent threat is not Xometry’s software but PCC-type primes’ suppliers automating while holding qualification, and primes keeping critical work in-house [inference].

Incumbents are partnering, not building: Lockheed embeds Hadrian [src-20261014-015327] and LM Ventures runs a $1B fund [src-20260401-073d24]; RTX Ventures led Radical [src-20250718-d46f24] and backs Fabri [src-20260617-4dfb96]; Siemens bought Altair and Dotmatics [src-20250326-cc1810][src-20250701-a7fef9]; Lindsay Goldberg is rolling up Alro [src-20260811-27702f].

Failures to learn from: - Markforged: $2.1B valuation 2021, sold to Stratasys for $42.5M in 2026 [src-20261023-63156a]. The same source notes Velo3D’s equity was wiped but the business recovered (Q1 2026 revenue $13.8M, up 48%) and that survivors were multi-process, not pure-play AM [src-20261023-63156a]. - Velo3D: defense refocus after forbearance and rescue [src-20261023-9f720f]; Q1 2026 GM 17.2% vs -73.6% [src-20261023-63156a]. - Bright Machines (failure example only): $1.6B SPAC fell through; Israel R&D closed with ~60 layoffs in 2024; sold $71.5M of a $145.5M 2022 Form D [src-20240527-6a4884]. - Daedalus: AI-run precision CNC factory (Karlsruhe), $21M Series A Feb 2024, insolvency Aug 2026 [src-20260801-4532f6]. The closest failed analog to the owner-operator thesis. - VulcanForms: founder CEO replaced, then $220M in Jan 2026 [src-20240919-72efcc][src-20260130-365541]. Pivot, not failure. - Velo3D debt cut 60% to ~$10M via debt-to-equity, Mar 2026 [src-20260311-79b04d]. - Lesson: survivors found a repeat defense buyer before scaling capex; equipment sellers did not compound [inference].

8. Customers and demand
In one line: buyers pay a premium for speed on urgent and prototype parts; evidence of repeat high-volume orders is still thin.
Who buys: primes (Lockheed, RTX, Northrop, Boeing, L3Harris, GA-ASI), new primes (Anduril), depots and government programs (Army, Navy, DIU), commercial hard-tech (SendCutSend's 60% of Fortune 500) [src-20261020-487a39].
What they value, side by side (all AlphaSense-arranged expert calls, med):

| Buyer | Price vs incumbents | Speed | Share of wallet / switching | |—|—|—|—| | Raytheon senior manager [src-20250708-2aa3ad] | Hadrian ~50% lower on RFQ quotes; no parts received yet; introduced via RTX Ventures | ~1/3 the lead time | Sweet spot $300-700 parts, no castings; suppliers at 80-90% utilization | | Boeing head of product [src-20260802-ef6506] | 20-25% gap would trigger re-sourcing | Hadrian 2-3x faster | Hadrian 2-3% of precision spend, prototypes and non-critical parts | | Boeing executive director [src-20260908-ed94f0] | At 1,000 parts a traditional shop is cheaper | Full inspection time equal | Losing Hadrian = minimal impact | | Northrop process engineer [src-20260802-36d46c] | Would pay 20-40% more to avoid program delay | n/a | Hadrian <10% of outsourced work, maybe 10-20%; CNC only; replacing it takes months to a year | | FlyX CEO, ex-

customer [src-20260806-3a2ec7]	Churned on cost; premium “10%-15% is tops”, 30-40% “not justifiable”	Speed “two times” a traditional shop; rework “one in thousands”	AS9100 Hadrian is “overkill” for unmanned work; aerospace customers “cannot afford to switch” once qualified	
Raytheon senior mech. engineer [src-20260823-b3713f]	n/a	n/a	Buys weekly but spend is “currently in the tens of thousands” in a probationary period; “multiple millions” possible later	
ICT USA / Wistron [src-20260805-59d4fe]	Discounted: the expert describes buying CNC machines, not parts, so the call is not buyer evidence	n/a	Only usable point: ODM keeps ~95% in-house, outsources low-volume aero/defense overflow	
Boeing CTO-org manager [src-20260818-9b2d5c]	Paid more so far; 5-7% discount is exciting, 30-50% would move volume			

n/a | No certified flying aircraft uses Hadrian parts | | Boeing strategy head [src-20250712-e5e110] | 20-30% per-part saving “optimistic”; expensive at low volume | 24/7 faster | Xometry better for low volume | | Northrop director [src-20250707-3b7a46] | Incumbents keep a cost edge, will price-war | Lead time is the pain | Rates Hadrian 8/10 | | Ex-Anduril hardware lead [src-20250709-43486e] | Defense buyers are price-insensitive | n/a | No production volume given | | Thales Alenia CPO [src-20251110-84b67e] | n/a | Capacity is the pain; Chrome VI banned from end-2024 | Prime does not qualify Tier-2 network sites; would only use suppliers with $30-40M revenue; fits smaller defense, auto and energy first | | Anduril senior engineer [src-20250623-5bc67c] | Wants conforming parts at competitive price | Lower lead times | Skeptical of “AI bells and

whistles”		Ex-Xometry VP [src-20260808-dc5cbb]	n/a	n/a	50-60% of parts arrive with no 3D model; CEO in every deal, so feast or famine	
		Defense startup chief of staff [src-20250729-a0a78c]	n/a	Atomic 8 days, Hadrian 8-9 days	Big buyers will move in-house after a “honeymoon period”	
		GE program lead [src-20250709-0250ae] (low)	Broker markup “pretty horrific”	n/a	Hadrian “too new, too young”	

The contradiction to put on the slide: Raytheon reports Hadrian *cheaper*; Boeing reports it *dearer at volume*. Both can be true if Hadrian prices aggressively on low-volume, urgent work [inference]. The demand that is proven is speed on prototypes and LRIP; demand at rate production is not yet proven [inference].

Qualification as moat or tax. Supplier qualification 9-12 months (Raytheon) [src-20250708-2aa3ad]; onboarding 1-2 years (Northrop) [src-20260802-36d46c]; Power says AS9100-level trust takes 7-8 years [src-20260908-65fab6] (claim). Airbus: supply base consolidating to fewer suppliers; networks suit prototypes [src-20251106-dfddcb]. A Raytheon engineer says quality from new suppliers now surpasses legacy [src-20260823-b3713f]. Read: a tax at entry, a moat once paid; Amca buys it via acquisitions [src-20260520-af3e6c] [inference].

Repeat orders (federal data). Firestorm is the only producer with a SBIR (Small Business Innovation Research, staged federal R&D grants) to Phase III IDIQ to repeat task orders ladder: 19 awards, $53.8M [src-20251022-629fc0][src-20251219-bd5788]. Hadrian’s Army award was $39.2M obligated, $80M total, one bid received [src-20260317-5585be]; the USAspending record ($79.4M) is described as “DATA RIGHTS AND INTELLECTUAL PROPERTY” [src-20260317-77e798], so on its face it is not a clean repeat parts-production order. Raytheon spend is “tens of thousands” so far [src-20260823-b3713f]. HII signed performance-based agreements that pay Path Robotics and GrayMatter on production output [src-20260101-164f12], a prime buying automation per unit rather than per robot (vendors, not producers). Mariana, Arda, 1872, Nox: no

prime awards found (insights/repeat-orders-evidence.md).

9. Capital and financing

In one line: venture money builds the first plant; loans and government money build the rest.

Company	Equity raised	Last round / valuation	Debt or gov	Capex datapoint	Who funds plant	Source
Hadrian	~$2.24B [Harmonic]	$1.37B D at $7.87B, Aug 2026	$360M revolver; Navy $900M	~$200M per 270K sq ft plant; ~$1B ahead of contracts (claim)	Debt, per Power: “equity never funds factory construction”	[src-20260807-b0dc5b][src-20260817-c48f28][src-20260320-67ba4c][src-20250820-cfbf02][src-20260921-876080]
Divergent	n/d; Series D $230M (Hexagon)	$290M E at $2.3B, Sep 2025	$40M debt in E	Long Beach 400-430K sq ft, cost n/d	Equity + debt	[src-20250915-311606][src-20231115-5d0589][src-20261023-e02543]
Isembard	$59M	$50M A, val. n/d	Undisclosed debt for owned plants	n/d	Franchisees fund their own plants	[src-20260310-45bab3][src-20250929-857f17]
SendCutSend	$116M + Oct round	$2.1B B (size n/d)	Equipment loans	~$50M new facility; first $750K laser financed	Equipment debt	[src-20261020-487a39][src-20260519-2d6ceb][src-20260227-e92d45]
Vulcan Elements	$65M A + $50M Commerce equity	n/d	$620M OSC loan; $550M private	10,000 t plant	Government-led	[src-20251103-ee9b27][src-20261010-8e6ec0]
Mariana	~$400M incl. project capital	$310M B, val. n/d ($1.5B Harmonic, low)	Project capital	n/d	Project finance	[src-20260803-30e6d5]

Pattern: equity for software, team and plant #1; banks, OSC and project finance for plants #2+ [inference]. Bear: Lowercarbon says modular-hardware companies at Series B often have ~$10M revenue at break-even gross margin [src-20251211-87dc10]; a six-machine first plant cost >$10M [src-20260817-8867f2]. ABL (asset-based lending against receivables and inventory) is not evidenced for any tracked startup (gap).

9A. Unit economics by layer

Unit economics means how much a company earns or loses on each part or ton it sells, after direct costs.

Startup margins are not disclosed anywhere in the repo. The table puts incumbent benchmarks next to the cost drivers the expert calls quantify. Cells marked (claim) come from a single speaker.

Layer	Gross margin benchmark	Capex and fixed cost	Variable cost and pricing	What moves the margin [inference]
metals-supply	Service centers 17.7% (Ryerson) to 31.6% FIFO (Reliance) [src-20260729-407fd7][src-20261021-16bb64]; Nucor 12% FY25 [src-20260225-3ea941]	Vulcan 10,000 t magnet plant on a $1.4B package [src-20251103-ee9b27]	Metal price passes through; tariffs lift inventory gains [src-20260728-ae7dea]	Quote speed and remnant use; price cycle dominates
fabrication	SendCutSend profitable 8 years, margin n/d [src-20260519-38c7ed]	~$50M per new facility; first laser $750K, debt-financed [src-20260519-2d6ceb][src-20260227-e92d45]	Xometry material can be 85% of a part's price (claim) [src-20260611-7b80f0]	Utilization (80-90% target) [src-20260626-541ac8]
precision-parts	Proto Labs 44.5%; own factories 49% vs network 31% [src-20260206-8bb619][src-20260206-2b13c7]; job shops 15-25% (thesis.md)	Plant 1: six machines, >$10M; Grob/Hermle 5-axis about $1M each vs Doosan $200K; target 20-25% low-cost machines (claim) [src-20260817-8867f2]; Mesa ~$200M [src-20250820-cfbf02]	Labor hours 100 to 30-50 per part, net saving 20-30% at best after capex (claim) [src-20250712-e5e110]; NRE $1-5K, casting tooling tens to hundreds of $K [src-20260611-7b80f0]; CAM staff $130-150K [src-20260817-8867f2]	Machining is 1/3 of the work; inspection, paperwork and certs are the rest [src-20250707-3b7a46][src-20260611-7b80f0]
materials-rd	Custom programs 5-15%, platform 85-90% (claim) [src-20261016-29997e]	Self-driving lab about $20M to build, low running cost (claim) [src-20260909-af3e60]	Buyer pays about $150K per software benchmark, $500K per pilot, $2M per validated candidate, $10-15M a year per lab line (claim) [src-20260909-af3e60]; ACVs "hundreds of thousands to millions" [src-20260913-3ad55e]	Hit rate and repeat campaigns; royalties rejected
factory-ai	Rockwell 49.5% [src-20260804-0e8130]	Software; needs a host plant	Arda usage-based [src-20261020-a1e174]; AI Robotics welding inspection on 24 lines, revenue up about 18x, profitable without outside capital [src-20261011-a8b14d]	False-detection rate on live lines [src-20261011-a8b14d]

Power and compute are minor costs: 1,200A service with 500-600A used for 6-8 machines; about 30 CNCs run on 2,000A (claim) [src-20260817-8867f2].

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10. Policy tracker (newest first)

The plain-English guide to each tool is in section 1A.4. This table is the dated list.

Date	Policy	What it does	Layer	Source
2026-10-20	Army 1M drones plan (reported)	Volume demand for magnets, airframes	metals-supply, fabrication	[src-20261020-878aa9]

Date	Policy	What it does	Layer	Source
2026-10-20	45X	Advanced manufacturing production credit	metals-supply	[src-20261020-7d90fa]
2026-09-23	SkyFoundry / Vulcan award	Army drone magnet supply	metals-supply	[src-20260923-c53c99]
2026-09-08	OSC	DoD lender for critical-tech capex	all producers	[src-20260908-f6558e]
2026-10-19	DPA Title III tally	23 awards, $820.5M since Jan 2025; rare earths $405.1M, solid rocket motors $196.4M; none to tracked startups	metals-supply	[src-20261019-68fd3d] (med, third-party)
2026-07-16	IEA public finance	Minerals commitments $65B	metals-supply	[src-20260716-6a06e4]
2026-04-06	Section 232 metals	50/25/10% structure; grid 15% to 2028	metals-supply, fabrication	[src-20260402-1b5d6c]
2026-01-01	DIU CSO	Commercial Solutions Opening, fast awards	precision-parts	[src-20260101-0b051b]
2025-12-18	FY26 NDAA (PL 119-60)	Sec. 225 robotic munitions automation; Sec. 867 Industrial Base Fund (castings, forgings, machine tools); Sec. 804(b) multiyear procurement from FY2026 for SM-6, Tomahawk, AMRAAM, JASSM, LRASM, PAC-3 and affordable mass munitions; Sec. 867 expands the Industrial Base Fund (castings, forgings, machine tools) and adds grants, OTAs and a purchase-commitment program of up to 10 years including advance payments; Sec. 1842 makes each program’s product support manager assess within 120 days which critical spares advanced manufacturing can make within 24 months, with NRE and who pays; Sec. 1652 Golden Dome cost reports; Sec. 1841 Civil Reserve Manufacturing Network; Sec. 1847 surge report	precision-parts, metals-supply	[src-20251218-f0d6ef] [src-20251218-6c4228]
2025-12-15	DIB strategy / DPA	DPA Title III (Defense Production Act authority to fund domestic capacity)	metals-supply	[src-20251215-98f650]
2025-11-03	OSC loan to Vulcan	$620M	metals-supply	[src-20251103-ee9b27]

Date	Policy	What it does	Layer	Source
2025-08-06	Apple-MP magnet commitment	Apple commits to buy US-made rare earth magnets from MP (no prepay or dollar figure in this source)	metals-supply	[src-20250806-a0a34e]
2025-08-01	Section 232 copper	50% on semi-finished copper content; refined copper 15% (2027) / 30% (2028) under review	metals-supply	[src-20250805-80f2f4]
2024-12-19	OTA authority	Prototype to production without recompete	precision-parts	[src-20241219-9f8fcc]

Gaps: Golden Dome dollar flows to tracked names. Sec. 1652 is reporting-only, but the act’s funding tables reference funds “supporting the Golden Dome missile defense system”; the amount is unconfirmed [src-20251218-f0d6ef]; no DPA Title III award to a tracked startup in the third-party tally [src-20261019-68fd3d]; OSC loan to Vulcan is the only federal loan to a tracked name [src-20251103-ee9b27].

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11. Companies worth attention

Bucket	Company	Layer	Model	Why	Next step
Target	Senra	fabrication	owner-operator	Series B size, real customers, in-house software	Founder call; FPY and repeat-order data
Target	Freeform	precision-parts	owner-operator	Best evidenced loop; $179M post	Customer reference calls (Boeing, DIU)
Target	Isembard	precision-parts	hybrid	Capital-light Hadrian variant; val. undisclosed; Thales CPO skeptical [src-20251110-84b67e]	Unit economics per franchise; site-audit answer for primes
Candidate	Fabri	precision-parts	owner-operator	Castings are ~40% of munitions (claim, Power) [src-20260921-876080]; RTX/LMT backed	Promote file; 2nd source
Candidate	1872	fabrication	owner-operator	Cheap entry, energy skids	Track to first repeat order
Candidate	Atomic Industries	fabrication	owner-operator	Tool and die bottleneck	Second source
Candidate	Machina	fabrication	hybrid	Depot demand; long Form D	Verify valuation
Benchmark	Hadrian, Divergent, SendCutSend, Mariana, Amca	various	various	Priced $1B-$7.9B	Use for comps
Benchmark	Periodic, CuspAI, Mind, Arda	materials-rd, factory-ai	various	Priced pre-product	Watch for revenue

Bucket	Company	Layer	Model	Why	Next step
White space	Service-center software for mid-size distributors	metals-supply	software-first	~9,400 establishments, top 50 hold ~50% [src-20251201-4b732d]; Nox passed	Sweep
White space	Forgings	precision-parts	owner-operator	DoD flags castings and forgings barriers [src-20220215-45c353]; no AI-native forger in repo	Sweep
White space	Inspection/NDT as a service	precision-parts	enabler	Inspection time equal for Hadrian vs legacy [src-20260908-ed94f0]	Sweep

12. Bottom line

Five conclusions 1. The category is real, but proven demand is speed on prototype and low-rate work; rate production share is 2-10% at the primes we spoke to [src-20260802-ef6506][src-20260802-36d46c]. 2. The data loop is evidenced (score 2) only where the process is continuous: Freeform, Mariana, Radical, plus Hadrian's training compression. Nobody shows improvement over time (section 6). 3. Capital splits into three tiers: capital-light owner-operators (SendCutSend, Senra), capital-heavy owner-operators financed by debt and government (Hadrian, Vulcan, Mariana), and software-first vendors that need a host plant (Arda, Mind) [inference]. 4. Precision-parts is priced out at the top (Hadrian $7.87B [src-20260807-b0dc5b], Amca $1B+ [src-20260520-af3e6c]); SPC entry is Series B in fabrication or a hybrid/franchise variant [inference]. 5. Policy pull lands mostly on metals-supply (OSC, Section 232 incl. copper, IEA finance, DPA Title III to rare earths and rocket motors) [src-20251103-ee9b27][src-20260402-1b5d6c][src-20250805-80f2f4][src-20261019-68fd3d]; the FY26 NDAA adds robotic munitions automation and an Industrial Base Fund for castings, forgings and machine tools, but no money to tracked startups yet [src-20251218-f0d6ef]; precision-parts gets pull through prime contracts instead [inference].

Layer x model matrix (our pick per cell, [inference])

	owner-operator	hybrid	software-first	enabler
metals-supply	Vulcan, Mariana (benchmarks)	n/a	White space (service-center software)	n/a
fabrication	**Senra** (pick); SendCutSend (benchmark); 1872	Machina	n/a	Path Robotics
precision-parts	**Freeform** (pick); Fabri	**Isembard** (pick); Hadrian, Divergent (benchmarks)	n/a	n/a
materials-rd	Radical, Periodic	n/a	CuspAI	n/a
factory-ai	n/a	n/a	Arda, Mind (benchmarks)	Siemens, Rockwell (incumbents)

Three targets: Senra, Freeform, Isembard (reasons in section 11).

Three risks 1. Rate-production economics: at 1,000 parts a traditional shop is cheaper (Boeing) [src-20260908-ed94f0]. 2. Capex ahead of contracts: ~$1B needed per Power [src-20260921-876080]; a munitions slowdown strands plants [inference]. 3. Valuation outrunning revenue: SendCutSend's

multiple is unknown (“nine digits” revenue at $2.1B) [src-20261020-487a39]; Periodic and Mind priced before product.

Strongest bear case - Rate economics: Boeing buyers report paying more, not less, and cost only matters after quality and delivery are proven [src-20260818-9b2d5c][src-20260908-ed94f0]; 2-10% share of wallet is prototype work [src-20260802-ef6506][src-20260802-36d46c]. - The model has failed recently: Daedalus insolvent [src-20260801-4532f6]; Bright Machines’ SPAC fell through [src-20240527-6a4884]; Markforged sold for $42.5M after a $2.1B valuation, and Velo3D’s equity was wiped though the business survived [src-20261023-63156a]. - Quality at complexity is unproven: ex-Anduril lead says Hadrian struggled on complex parts [src-20250709-43486e]; Thales will not qualify a franchise network [src-20251110-84b67e]. - The AI is thinner than the pitch: Opus rides third-party CAM [src-20261004-9b47f5]; robot learning at ~60% vs 99.9% [src-20260904-267293]; A-Lab’s 41 novel compounds challenged by an outside critique (not retracted) [src-20240124-513fd9]. - Valuation: Hadrian $7.87B, Mind $3.4B, Periodic ~$7.5B (reported) with no audited revenue; a defense-budget or tariff reversal hits capex built ahead of contracts [src-20260921-876080]. - Slack exists: 34% spindle utilization in incumbent shops [src-20250120-419768] means demand can be met without new plants. - Share per customer is capped by policy: a 1/3-of-revenue cap at SpaceX [src-20250708-342eaf], no single-sourcing after the PCC fire [src-20260818-9b2d5c], and 5-10 years to trust a new supplier [src-20250712-e5e110]. - Regulators get stricter after Boeing’s quality lapses: “Innovation … is the first thing that gets sacrificed, and Hadrian falls in that innovation category” [src-20260818-9b2d5c]. - Team durability: 90% of Hadrian’s first 10 employees gone by year 5; a 40-person software team churns [src-20260817-8867f2]. - AI CAM vendors sell the same automation to every incumbent shop: CloudNC CAM Assist is in GibbsCAM [src-20261021-9769d1][src-20261009-7cafd0]. Ex-Hadrian staff call CAM vendors the main threat [src-20250708-342eaf]. - Some buyers see financing, not venture: “better suited to debt than VC” [src-20250709-43486e]; “propped up by VC money” (peer view) [src-20260808-dc5cbb]. - Materials-rd: model leadership turns over on a “three-month cadence”, so stickiness is falling [src-20260909-af3e60] (low); chemical customers have “25 data points”, not big data [src-20260913-3ad55e].

Bias check on targets. Senra (10 sources), Freeform (14) and Isembard (13) are among the least-covered names, not the most (Hadrian 76, Machina 42, Arda 33, SendCutSend 31). That avoids picking for press volume but means each target rests mostly on company claims; Isembard also has the one negative buyer call. Treat all three as hypotheses for diligence, not conclusions.

What would change our mind - A prime shows Hadrian or peers at >20% of rate-production spend with lower unit price (bull). - A company publishes scrap, cycle time or quote accuracy improving with volume (moves loop to 3). - Section 232 struck down or rolled back (bear for metals-supply and fabrication). - A tracked owner-operator raises a down round or stalls a plant (bear).

Appendix A. Glossary

Term	Definition
A&D	Aerospace and defense
ABL	Asset-based lending: credit secured by receivables and inventory
ACV	Annual contract value
AD/CVD	Antidumping and countervailing duties on unfairly priced imports
AM	Additive manufacturing: 3D printing
AS9100	Aerospace quality management standard; table stakes for flight parts

Term	Definition
AS9102	Standard form for a first article inspection report
B2B / B2G	Selling to businesses / to government
BOM	Bill of materials
CAM	Software turning a 3D model into machine toolpaths
Capex	Capital expenditure: money spent on plants and machines
CHIPS Act	2022 law funding US chip factories ($52.7B)
CMMC	DoD cybersecurity certification for contractors
CNC	Computer numerical control: a computer-driven cutting machine
CRO	Contract research organization: an outside lab hired for experiments
CSO	Commercial Solutions Opening, a fast DoD solicitation
DFARS	DoD procurement rules, incl. specialty-metal sourcing
DIB	Defense industrial base: every company that supplies the Defense Department
DIU	Defense Innovation Unit, which buys commercial tech fast
DoD / DoW	Department of Defense, also called Department of War since 2025
DPA Title III	Defense Production Act authority to fund domestic capacity
EBITDA	Earnings before interest, tax, depreciation and amortization; a rough cash-profit measure
FAI	First article inspection: full check of the first part made
FFP	Firm fixed price contract
Form D	SEC filing a company makes when it sells private shares
FPY	First-pass yield: share passing inspection first time
GM	Gross margin: revenue minus direct cost, as a share of revenue
IDIQ	Indefinite-delivery, indefinite-quantity contract vehicle
IEA	International Energy Agency
IRA / 45X	Inflation Reduction Act (2022); 45X is its per-unit manufacturing tax credit
ITAR	US export control limiting defense work to US persons
Job shop	Small owner-run shop building to a customer’s drawing
LIFO/FIFO	Inventory accounting methods; LIFO lowers margins when prices rise
LPBF	Laser powder bed fusion, a metal 3D printing process
LRIP	Low-rate initial production
MES	Software that tracks and directs shop-floor work
NADCAP	Accreditation for special processes (heat treat, NDT, coatings)
NDAA	National Defense Authorization Act, the yearly defense policy law
NDT	Non-destructive testing
Nesting	Packing parts on sheet or stock to minimize scrap
NRE	Non-recurring engineering: one-time setup cost before production

Term	Definition
ODM	Original design manufacturer: builds products others sell under their brand
OSC	DoD Office of Strategic Capital, a lender
OTA	Other Transaction Agreement, fast non-FAR prototype contract
PPBE	DoD planning, programming, budgeting and execution process
Prime	Top contractor selling whole systems to the government (Lockheed, RTX, Boeing)
RFQ	Request for quote
RL	Reinforcement learning: AI that learns by trial and reward
SBIR	Staged federal R&D grants to small businesses
Section 232	National-security tariff authority (1962 statute)
Service center	Metal distributor that stocks, cuts and ships
SPAC	A shell company that lists a private company on the stock market
TAM	Total addressable market
Tier-1/2/3	Supplier levels below a prime: sub-systems, then parts, then raw inputs
USAspending	Public federal database of contract awards
Utilization	Share of available machine hours actually producing

Appendix B. Gaps and next actions

- Hadrian revenue: two low-confidence figures conflict; get a primary number.
- Startup gross margins: none disclosed; ask Senra, SendCutSend.
- Refining TAM and economics: no free source found.
- Fabrication TAM: none; Census CBP gives counts and payroll only [src-20250626-23f6d9]. Next: 2022 Economic Census receipts by NAICS 3323/3327.
- AI/robotics cost decline over 24 months: no source.
- Golden Dome amounts: the NDAA funding tables reference Golden Dome funds; size and recipients unconfirmed [src-20251218-f0d6ef]. Freeform Skyfall go-live: unconfirmed.
- ABL use by startups: none evidenced.
- Exits (Q14): no comparable acquisition above $100M in repo.

Appendix C. Source coverage (ledger, 2026-10-25)

Per-source ledger: `insights/exports/primer-ledger.csv`; reading notes kept outside the PDF.

Bucket	Read status
Expert calls (all 28 on-thesis)	Read in full, line by line
Podcasts	40 of 102 transcripts read in full (the top 20 by Lilly’s choice plus earlier passes). The other 62 (about 485K words) are not yet read in full; a full read is queued in 12 batches and its findings will be added in the next version
Sources the primer cites (~200)	Read in full, except: the FY26 NDAA (about 517K words) checked against every primer claim plus the manufacturing, munitions, reshoring and tariff sections; the DIU guidebook and the boilerplate parts of 10-Q/10-K filings (e.g. ATI) skimmed

Bucket	Read status
On-thesis sources not cited (~390)	Read in full; 99 turned out to be a title or placeholder only. One 589K-character academic review of self-driving labs not read (general background)
No real text captured	Fast Radius (SEC block page), VulcanForms layoffs, Siemens (menu JSON) and other stubs: claims citing them were removed
Form D / award stubs, noise, duplicates	Data rows checked; category confirmed by keyword scan, not read line by line

Honest limits: 62 podcast transcripts and the long filings above were not read line by line. Funding and valuation discrepancies were dropped by design.